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Data analytics for Cooperative Intelligent Transport Systems

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ABSTRACT

Cooperative Intelligent Transport System (C-ITS) is a key enabler of future road traffic management systems. The core component of C-ITS includes vehicles, road side units and traffic command centers. They generate a large amount of traffic that is made up of both mobility and service-related data. To extract useful and relevant information out of this data, data analytics will play a crucial role in future C-ITS applications. We present a review of how data analytics can benefit C-ITS applications. We describe the various types of data generated by C-ITS applications and potential dissemination techniques using various wireless technologies. We demonstrate how meaningful results from this data could be beneficial to C-ITS. We also demonstrate the improved reliability of C-ITS applications that can be achieved with data analytics using simulation results. Finally, we discuss future possible applications of data analytics in C-ITS.

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1. Introduction

Cooperative Intelligent Transport System (C-ITS) is a promising technology that aims to improve road safety, traffic management and driver comfort. It will be an integral component in the future development of smart cities. The concept behind C-ITS is ubiquitous connectivity of vehicles to provide them with a good knowledge of the traffic conditions on the road [1–3]. Along with vehicle to vehicle connectivity, infrastructure Road Side Units (RSUs) are also placed at various geographical locations to disseminate data from vehicles to central Traffic Command Centers (TCCs). The centralized TCC helps in managing the city level traffic, ensuring efficient emergency warning notifications [4] and analysis of traffic related data for efficient route evaluation [5]. The TCC also provides the wireless transceivers of vehicles with important information for congestion control [6] and helps in the selection of security algorithms [1].

C-ITS deals with a large amount of data exchange between different C-ITS entities namely, vehicles, RSUs and TCCs. Data analytics is a key tool that can be used to extract useful results out of this data. Additionally, data could also be obtained from various other sensors on the road and cellular phones to support C-ITS applications. The goal of data analytics in this context is the accurate interpretation of data and making accurate deductions to optimize the operation of C-ITS which will enhance the reliability and robustness of C-ITS applications.

A key challenge that arises when data analytics is applied to C-ITS is data generation and transmission. Since C-ITS includes various applications, the data requirements for each of them needs to be clearly defined. Data analytics can yield meaningful results if the data quality is high. C-ITS standards have come up with a set of messages and data that needs to be exchanged among C-ITS entities along with their communication requirements [7], [8]. However, these requirements do not remain static and need to be varied according to many factors such as nature of application, current network load, and so on. Moreover, the data requirements for many emerging applications that utilize C-ITS are yet to be defined.

C-ITS applications generate a lot of data which needs to be collected and sent to the TCC for data analysis [9]. IEEE 802.11p is the default wireless technology that is used for efficient data dissemination in C-ITS. LTE-V2X is another potential wireless technology that can play a critical role in the ubiquitous dissemination of traffic and mobility data [3], [10]. It is vital to select the most suitable wireless technology for a particular application. This wireless technology should also work in cooperation of heterogeneous wireless networks.

Since vehicles traveling on the road in C-ITS are connected in an ad hoc manner, decision making for safety applications could occur

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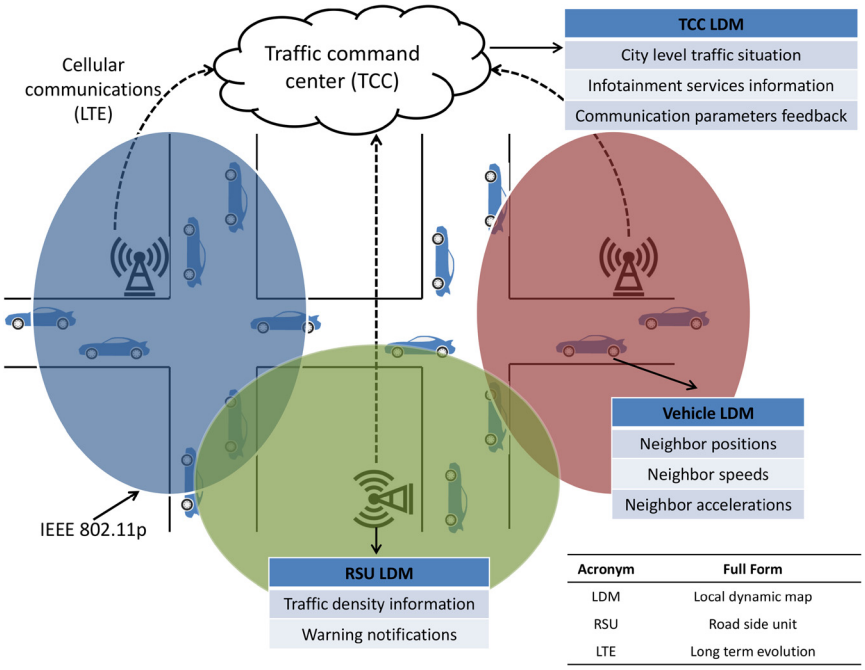


Fig. 1. LDM hierarchy.

in a distributed manner. In contrast, traffic management applications or non-safety applications could make use of a centralized approach where the TCC makes the decision. Thus, the TCC could include a data analytics module to control various aspects of C-ITS communications. The results from the data analytics module provides feedback to various C-ITS stations on how to adapt their parameters in order to maximize communication efficiency.

The use of data analytics provides a solution to address critical challenges in C-ITS. For example, Quality of Service (QoS) of various applications can be improved using traffic density on the road [11]. Long term analysis of traffic mobility data can help travelers to make decisions about efficient routes to the destination and also help to better manage the overall city traffic.

1.1. Main contributions of this work

We summarize the main contributions of this work as follows:

- We discuss different challenges related to applying data analytics in C-ITS.
- We demonstrate how data analytics could benefit C-ITS using two application scenarios (including efficient route guidance and adaptive transmission range) along with an evaluation of their performance using extensive simulation results.
- We discuss future research directions on applying data analytics in C-ITS.

2. Related works

In this section, we describe the C-ITS architecture and we review recent related works on data analytics in C-ITS.

2.1. C-ITS architecture

The core component of C-ITS is its communication nodes known as ITS entities that include vehicles, RSUs and TCC as shown in Fig. 1. While vehicles are mobile ITS entities whose movement is determined by road topology, RSUs and TCC are the static entities of C-ITS assisting applications to maintain wide area connectivity. Each vehicle is equipped with a wireless transceiver known as the On-Board Unit (OBU) which provides communication capabilities to the vehicle.

Vehicles interact with each other and RSUs to enable applications for cooperative awareness and traffic management. In this context, RSUs help to extend the communication range of vehicles by relaying notifications to a wider geographical area. RSUs also help in the centralized evaluation of key C-ITS parameters such as current traffic density on the road, average vehicle speeds and vehicle traffic flow. Moreover, an RSU can also act as relay between vehicles and the TCC, the latter taking the role of service provider and city-wide traffic manager. TCC can monitor the traffic situation of the whole city and provide information regarding available infotainment services. In addition, TCC can also provide feedback to vehicles on how to select their communication parameters such as transmission range, data rate and packet generation rate in order to maximize the packet transmission reliability. Since TCC has a global view of the whole network, it can make accurate selection of these parameters to enhance the overall communication efficiency of the network.

At the heart of C-ITS is a database known as Local Dynamic Map (LDM) which is maintained with the help of each of the above C-ITS entities. The map is central to the operation of C-ITS applications because it maintains an updated view of traffic mobility and information about road traffic conditions. In fact, LDM is a conceptual data store embedded within each C-ITS entity which takes input from different sources such as vehicles, RSUs and TCC as shown in Fig. 2.

The data stored in LDM can be of four types [12]. The type 1 data is permanent static data obtained from map information such as road topology, road speed limits and point of interests. The type 2 data is transient static data such as information about traffic signals and signs. Transient dynamic data including road works, weather conditions and changes in traffic conditions. This data is classified as type 3. Finally, type 4 data is highly dynamic information provided by Cooperative Awareness Messages (CAMs) and De-centralized Environmental Notification Messages (DENMs). CAMs are regular periodic messages that are exchanged among ve-

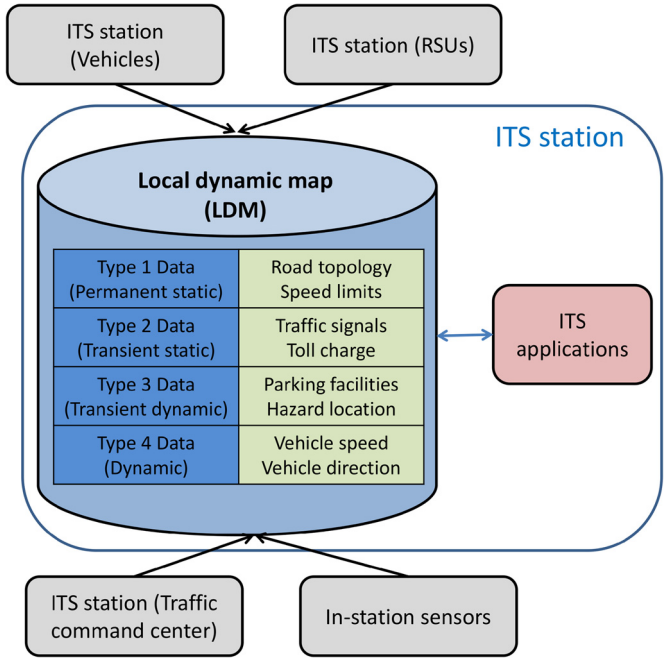


Fig. 2. Local dynamic map.

hicles to share mobility and traffic information whereas DENM are notification warnings that are issued during an emergency event.

Each C-ITS application utilizes relevant data from LDM. For example, a lane change application will take into account type 1 and type 4 data. Regarding type 1 data, the ETSI standard recommends not putting it in LDM of all C-ITS stations because a common map format for each application cannot be established. So, type 1 data can be stored as per requirements of the specific application such as navigation. Furthermore, the local digital mapping information could be used to reduce the storage complexity of LDM.

Since LDM is a key component of C-ITS, its accuracy is a vital requirement in ensuring the reliability (successful delivery of messages) of applications. An accurate awareness of the surrounding traffic helps vehicles take error-free driving decisions such as applying brakes and crossing an intersection. Without an updated LDM, vehicles could receive many false positives or false negatives, thus disrupting smooth traveling [1]. Moreover, an outdated LDM could affect city level traffic management decisions taken by the TCC. Data analytics plays a crucial role in improving the accuracy of LDM because it can provide a feedback on the optimal selection of communication parameters, thus maintaining a reliable data exchange between C-ITS stations.

2.2. Review of data analytics techniques in C-ITS

In C-ITS, data analytics techniques have been used to improve the reliability of communications in terms of congestion control and cooperative transmissions. There are many C-ITS applications that also rely on making intelligent decisions based on the data collected. Some data analytics techniques also leverage efficient distributed computing of big data in C-ITS. Next, we review recent research efforts in each of these categories and we summarize the key ideas in Table 1.

2.2.1. Congestion control

Since C-ITS requires different types of data (such as CAMs, warning notifications and other service messages) to be shared among vehicles and between vehicles and RSUs, they can cause data congestion particularly in a high network density scenario. Techniques in this category detect congestion based on collected data such as channel load, packet error rate and take measures to reduce congestion. In [13], the authors propose a congestion control strategy for intersection scenarios (where vehicles stop due to red light) with the assistance of RSUs. The congestion control mechanism makes use of three units namely, the congestion detection unit, the data control unit and the congestion control unit. In the congestion detection unit, when the channel usage level (percentage of time the channel is sensed as busy) value is greater than 70%, congestion is assumed. In the data control unit, the data of V2V communications near the intersection is gathered at a RSU and filtered to remove redundant transmissions. Using K-means clustering, the data is then clustered based on message size, message type, message validity time and the distance between vehicles and the RSU. Lastly, the congestion control unit adapts the communication parameters (such as transmission rate, transmission range and contention window size) of each cluster in order to reduce the end-to-end delay.

A fair transmit power control algorithm that can mitigate data congestion is presented in [14]. Each vehicle detects congestion using channel busy time metric and shares it with other vehicles in its proximity. A coalition is formed between vehicles within the carrier sensing range of each other. Shapley value is used to compute the marginal congestion contribution of each vehicle towards the coalition and vehicles therefore calculate transmit powers in a fair manner.

The work in [15] proposes an efficient queuing technique to reduce the processing time of the safety messages received. The processing time can be high when many messages are received at a vehicle and the available hardware has limited processing capability. The authors classify each received message into streams (based on message type and transmitter distance) and use the priority queue to process urgent information in order to keep the delay to a minimum.

2.2.2. Cooperative transmission

Cooperative transmission is a promising technique that increases the probability of packet reception using multiple relay vehicles [16]. In this technique, a key decision is which vehicle to choose as the relay and at what time that vehicle should transmit.

In [16], the authors proposed a cooperative MAC protocol (Time Division Multiple Access based) that improves data transmission reliability and network throughput. Each vehicle shares data such as its own time slot, one-hop neighbor list and time slots utilized by one-hop neighbors. In case of a direct transmission failure, a helper vehicle is selected as the one which has an available free time slot and is within one-hop from the destination.

A joint relay selection and power allocation technique for hybrid decode-amplify-forward network is presented in [17]. The channel statistical feature metric (calculated by using the variance of fading coefficients) is computed and the node with the maximum value of this metric is selected as the relay. In the next step, power allocation is done to meet the outage probability threshold which is computed using the data obtained from the channel model.

2.2.3. C-ITS applications

Many C-ITS applications collect and analyze sensors data and provide services to users based on it. Here, we discuss two such applications namely, smart parking system and road condition monitoring.

In [18], the authors propose a cloud-based smart parking system using Internet of Things (IoT). Each vehicle is equipped with a RFID tag and the parking space has a RFID reader at the entry and the exit points. The proposed system collects data about available parking spaces using RFID readers. Users can book a parking in ad-

Table 1
Review of data analytics techniques in C-ITS.

Category	Goal of the technique	Data gathered	Analytics technique used
Congestion control	Congestion control for intersection scenarios with the assistance of RSUs [13]	Channel usage level metric V2V messages	Threshold detection of channel usage level metric Calculating end-to-end delay for all combinations of communication parameters
	Fair transmit power control to reduce congestion [14]	Channel busy time metric	Shapley value algorithm
	Reduce processing time of received safety messages [15]	Safety messages	Priority queuing
Cooperative transmission	Cooperative MAC protocol to enhance network throughput [16]	List of one-hop neighbors Vehicle's utilized time slot Time slot utilized by neighbors	Direct transmission failure detection
	Joint relay selection and power allocation for hybrid decode-amplify-forward network [17]	Channel statiscal feature metric Channel model data	Maximum channel statistical feature metric Calculating outage probability
C-ITS applications	Smart parking system using IoT [18]	RFID tag data	Calculating available parking slots
	Crowdsensing based road condition monitoring [19]	Vehicle trajectory data	Wavelet packet-denoising and feature extraction Support vector machine
Distributed computing of big data	Hadoop framework to detect anomalous traffic events from large traffic data [20]	Traffic event data	MapReduce framework in Hadoop Parallel processing
	Data management for dynamic highway toll pricing [21]	Traffic data from sensors Toll revenues data	Spark MongoDB

vance using a mobile application. The decision to allocate parking is made by a central server.

A crowdsensing-based road condition monitoring system is presented in [19]. Vehicles are equipped with sensors such as accelerometers and gyroscopes. To collect data about road conditions, the authors conducted practical experiments using vehicles of different sizes and year of manufacture. Road anomalies are classified based on the trajectory data of these vehicles. At first, wavelet packet-denoising is used to reduce noise in the data. This is followed by applying feature extraction techniques to infer the actual road conditions based on received data. Furthermore, the Support Vector Machine (SVM) technique [22] enables the classification of road anomalies based on their severity.

2.2.4. Distributed computing of big data

As C-ITS generates large amounts of data, efficient parallel and distributed computing techniques are required. Hadoop and Spark are two commonly used tools that have been used for efficient distributed computing of big data in C-ITS.

In [20] the authors use Hadoop tool to analyze large amounts of traffic data. In particular, they use MapReduce framework in Hadoop to divide large scale traffic event data into sub-events. Parallel processing is then used on sub-events to derive anomalous traffic events.

In [21], the authors propose a data management technique for a dynamic highway toll pricing application. Spark [23] was used as a parallel processing technology to increase data processing speed. Spark was also used to perform data cleaning [24] and data harmonization (converting data from different sources into a uniform format for processing). MongoDB technology [25] was used for data management and storage. Traffic data from sensors and toll revenue data was used to estimate the optimal highway toll price.

3. Data analytics in C-ITS: challenges

In this section, we discuss some key challenges related to data analytics in the context of C-ITS. Table 2 highlights these challenges and their possible solutions.

3.1. Data collection

Every station in C-ITS collects data related to traffic and mobility information. Both the IEEE and the ETSI standards have allocated appropriate spectrum in the range of 5.9 GHz for exchanging C-ITS related data. The resulting spectrum known as Dedicated Short Range Communications (DSRC) is further divided into a control channel and various service channels for sharing safety and non-safety data messages respectively.

The ETSI standard has categorized safety data for C-ITS applications into two types namely, CAMs and DENMs [7]. Vehicles exchanging periodic messages in the form of CAMs are the major source of this safety data. The mobility and status information of a vehicle (such as its current location, speed and direction) are included as part of CAM. Each vehicle generates CAMs at a frequency of 1 to 10 packets per second. The communication range for such messages is around 150 to 1000 m. In contrast, warning notifications and long range data messages are handled by DENMs. These are event driven messages that are generated after an event has occurred. Using multiple hops, DENMs distribute notifications.

In addition to the above messages, different sensors on the road or outside buildings could collect other types of data such as available parking spots, detecting traffic congestion and so on [26]. RSUs could also use CAMs to determine the vehicle density in the surrounding and other important traffic metrics within a geographical region. Similarly, for infotainment applications, RSUs and TCC could collect data such as available parking spots in a mall, number of customers in a gas station and new sales at stores. Based on this data, they could provide services such as parking slot reservation, optimal gas station selection and advertisement of sales.

With the increasing number of safety and non-safety applications resulting in generation of big data in C-ITS, the sharing of all the information among vehicles and TCC could become computationally expensive. While vehicles could still share detailed traffic data within a certain geographical area, to improve network efficiency only the required information (such as safety awareness level, congestion level, traffic density level) should be propagated for long range communications. Therefore, another challenge is to determine the appropriate application parameters that could give meaningful insight into the traffic conditions and vehicle safety

Table 2
Data analytics challenges and solutions in C-ITS.

Challenges	Solutions
Data collection	CAMs DENMs Application-based parameters
Data dissemination	IEEE 802.11p LTE-V2X Bluetooth Zigbee
Data computation	Distributed (Vehicles) Centralized (TCC)
Metrics	Packet success rate Safety awareness
Data analysis	Correlation calculation
Feedback	Adaptation parameter Linear adaptation Sigmoid adaptation Fuzzy logic-based adaptation

(parameters such as safety awareness level [1], [9] can measure vehicle safety information in a single metric). By sharing these specific parameters instead of raw data, computational resources as well as the wireless spectrum could be more efficiently utilized.

3.2. Data dissemination

Another critical challenge related to data analytics is efficient data dissemination. As discussed earlier, for making decisions related to safety applications, vehicles share periodic CAMs. It is vital for vehicles to receive these messages with high reliability because safety decisions are made based on the neighborhood mobility information. In this context, IEEE 802.11p is the default wireless technology to disseminate these messages within the local neighborhood area. For long range messages such as warning notification, a multi-hop communication strategy could be employed [27]. Multi-hop communications could also be useful for communications between vehicles and RSUs.

For city level traffic management and other service applications, data needs to be disseminated to TCC. Long Term Evolution (LTE) is an effective wireless technology that can be used in conjunction with IEEE 802.11p for city wide connectivity [28]. The support for vehicular communications in LTE was included as part of the Third Generation Partnership Project (3GPP) release 14 (the standard is known as LTE-V or LTE-V2X) [10]. The key additions in LTE-V2X are improved reliability (achieved by redundant transmissions), and two radio interfaces namely, cellular and direct. The cellular interface supports V2I communications whereas the direct interface is used for Device to Device (D2D) communications [29]. The major advantage of LTE D2D communications [30] for disseminating CAMs is reduced end-to-end delay which makes it useful for vehicular safety applications with delay sensitive requirements. The most recent 5G V2X standard (release 15) by 3GPP introduces new enhancements that support use cases such as autonomous driving, sensor data dissemination and map sharing [10]. A more in-depth discussion on 5G technologies for vehicular communications is presented in [31].

3.3. Data computation

Data computation in ITS could be done either in a distributed or centralized manner. Since vehicles are connected in a distributed way and share regular CAMs, each vehicle can analyze this data and make its own safety decision such as applying brakes, changing lanes, and other operations [32]. For applications such as traffic management and warning notifications, centralized RSUs and TCC could cooperate to analyze geographical traffic data. This information can then be transferred to vehicles using a wireless dissemination technique.

3.4. Metrics

Another challenge in the use of data analytics for C-ITS is the use of efficient metrics for performance measurements. Evaluating the level of vehicle's awareness and knowledge of its surrounding is a crucial measure when evaluating the reliability of safety applications in a C-ITS. By maximizing this safety awareness metric, vehicles' safety could be ensured. However, how this safety awareness should be measured remains a challenging task. An accurate performance indicator for vehicle safety awareness is the accuracy of received information in CAMs and reliability of LDM. One of our recent works [9] addressed this challenge by proposing appropriate safety awareness metrics in C-ITS.

3.5. Data analysis

Another important aspect of data analytics is accurate data analysis. RSUs and TCC could receive lot of information such as vehicle density, average number of neighbor vehicles, average packet loss and key size of the security algorithm (a higher key size means more secure algorithm) used by vehicles in the network. Based on such data, appropriate inferences can be made. For example, a high packet loss and larger key size of the security algorithm could be correlated because a more secure algorithm may improve the overall level of security but actually reduces the number of packets delivered successfully in the network. Similarly, a high vehicle density may reduce safety awareness because the wireless channel could become congested.

3.6. Feedback

Another challenge when applying data analytics tools in C-ITS is the feedback provided to the vehicles to adapt their parameters based on the inferences made by the data analysis module. The feedback module needs to decide which parameter of the vehicle needs to be adapted i.e., application, wireless MAC/PHY, or traffic safety. In addition, the adaptive logic should be developed. The feedback module should take into account the QoS of applications and decide the corresponding logic i.e., linear adaptation, sigmoid based weighted adaptation, fuzzy logic rules that should be used. As an example, optimal transmit power control of wireless transceiver at vehicles could be decided based on the vehicle density. A sigmoid based weighted adaptation can cause a decrease in transmit power in order to efficiently use power control when the vehicle density increases beyond a certain threshold. Similarly, fuzzy logic rules could be used to make better driving decisions based on the position of critical neighbors in the vicinity.

4. Using data analytics for C-ITS applications

In this section, we present two example scenarios to show how data analytics can benefit C-ITS applications. The first scenario is an efficient route guidance application whereas the second scenario adapts vehicle's wireless transmission parameters to increase C-ITS application reliability.

4.1. Efficient route guidance application

Efficient route guidance is an important application of C-ITS that can help drivers save travel time. Current systems (such as Google maps, Navigation, Waze, and so on) rely on crowd-sensing

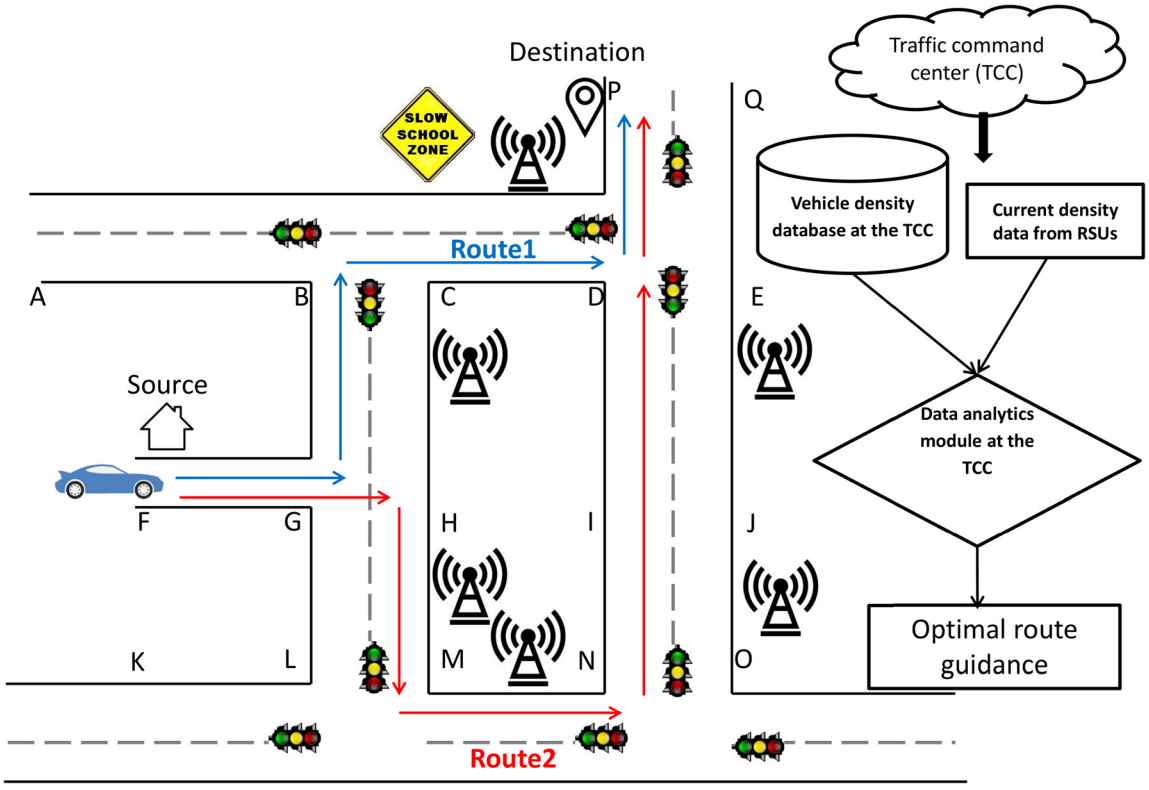


Fig. 3. Data analytics in efficient route guidance.

technology where users contribute (by sending their location updates on a regular basis or by manually inputing information about accidents, road closure, and so on) toward the estimation of traffic congestion. A major disadvantage of such systems is the non-availability of enough data from many roads (due to users not contributing by turning on their location information or manually inputting information). In contrast to these existing systems, C-ITS can provide real-time data on traffic conditions using sensors, road side units and on-board units [33]. This data includes information about accidents, traffic light status and traffic congestion. Therefore, C-ITS can provide more accurate and up-to-date information to users as compared to existing applications, thus allowing them to select faster routes to their destinations.

Fig. 3 depicts a route guidance application scenario. It shows an urban road topology where a traveler is planning to travel between his/her home (or source) and the destination point. There are two possible routes for this trip. The first option is to use road sections FG-GB-BD-DP (route1) to reach the destination. Although it is the shortest route, there is a school zone on the road section BD. The second option is to take the route FH-HM-MN-NP (route2) which although longer, there is no school zone in this path.

RSUs located on each road section provide wide area network connectivity and measure the traffic flow. Based on periodic sharing of CAMs by vehicles, RSUs can determine the number of vehicles present within a geographical area of around 500 to 1000 m. As discussed in Sec. 2.1, these RSUs further forward their vehicle density information to the centralized TCC using LTE. It is worth noting that the TCC saves this information in a database and uses it to evaluate long term road mobility statistics and city level traffic management.

When a traveler is planning to initiate his/her journey, he/she sends a query request to the TCC to receive information about the shortest route. TCC executes an algorithm that uses the vehicle density information received from RSUs on each road section to find the shortest path. To improve the route guidance decision, the TCC also uses the history of the vehicle density data for that road at that time of the day. This information is taken from the database of the TCC as shown in Fig. 3.

The estimated vehicle density on the road, represented by $\hat{\rho}$, can be calculated by using a weighted summation of current and past vehicle densities as shown in eq. (1). Here α is the weight assigned to the current vehicle density. The parameter α can be selected by the TCC based on the road and time of the day. For example, roads where the vehicle density variation is very high at a particular time, α can have a larger value to assign higher weight to the current vehicle density. In contrast, roads with little variations in vehicle density can allocate more weight to the past data and hence, a lower value of α can be chosen.

$$\hat{\rho} = \alpha\rho_{current} + (1-\alpha)\rho_{past} \tag{1}$$

We implemented the above route guidance application scenario using the NS-3 network simulator. Fig. 3 shows the road topology simulated. The algorithm for vehicle mobility traces was developed in MATLAB and then imported to NS-3. The vehicle density on each road section is varied from 50 vehicles/km to 200 vehicles/km. Each road has two lanes in each direction. The traffic signal cycle length is taken as 30*s*. However, the first traffic signal on the road section AD is adaptive and remains ON for up to 3 minutes when the school zone timings are in operation or there is an accident on the road. This is done to reduce traffic flow from the road section BG because the road section AD is already congested.

Vehicles broadcast periodic CAMs at a frequency of 10 times per second with a total size of 300 bytes including the security overhead. The transmission range and data rate are taken as 500 m and 6 Mbps respectively. The complete simulation parameters are given in Table 3 and are selected as per the recommendation of the ETSI standard [7].

The road scenario is simulated at four different times of the day: including the early morning time of 6am, the peak afternoon

Table 3
Simulation parameters.

Parameter		Value
Traffic	Road Area	2 km × 2 km
	Number of Lanes	4 (2 per direction)
	Vehicle Density	50 – 200 vehicles/km
	Traffic Signal Cycle Time	30 s–180 s
CAM	Transmission Range	500 m
	Generation Interval	100 ms
	Packet Size	300 bytes
	Data Rate	6 Mbps
Propagation Model	Pathloss	Dual-slope
	Fading	Nakagami $m = 1, 1.5, 3$

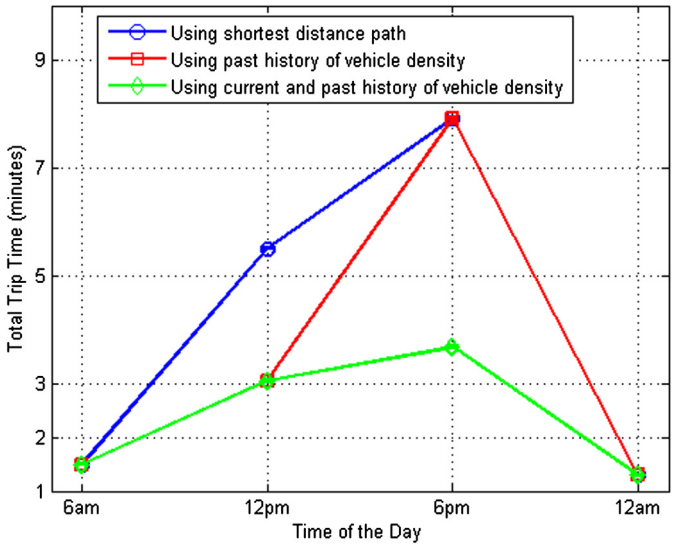


Fig. 4. Total trip time at different times of the day.

time of 12pm (when school zone time is in use), the evening time of 6pm (where an accident has also occurred on the road section AD) and late night time of 12am. Fig. 5 shows the route (either route1 or route2) chosen by the vehicle shown in Fig. 3 to travel from source to destination based on using the shortest distance path, and based on the result from the data analytics module of the TCC (as explained in this section). Furthermore, two techniques in the data analytics module are shown, one in which vehicle density at the TCC is estimated only based on the past history (using its database), and second where vehicle density calculation considers both the current data from RSUs and the past history.

We note that for the two times when there is very little congestion i.e., 6am and 12am, the shortest path calculation and the data analytics module choose the same path to route the vehicle from the source to the destination (i.e., route1). However, for the afternoon time of 12pm, the data analytics module estimates traffic congestion on route1 and recommends the alternate route for the trip. Hence, both versions of the data analytics module choose route2 as shown in Fig. 5.

For the evening time of 6pm, we have simulated an accident event on the road section AD. As a result, traffic congestion starts building up along the shortest route at the same time the vehicle is planning its trip. This event is a sudden event and past history does not show a traffic congestion on this road section at this particular time. In this case, if we use only the TCC database of past vehicle density, route1 will be chosen. However, if we use the current vehicle density information in conjunction with the past history, TCC recommends choosing route2 to avoid the unexpected congestion.

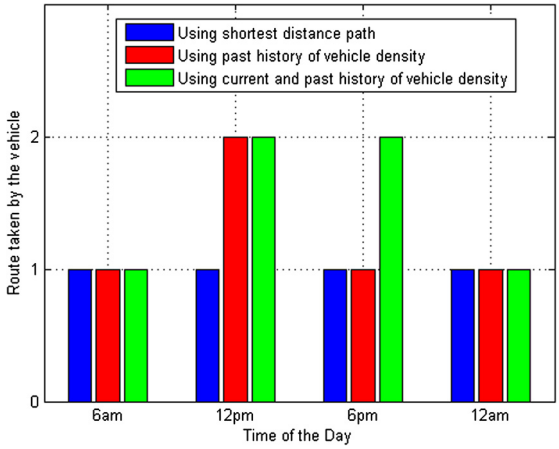


Fig. 5. Route taken by the vehicle at different time of the day. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Fig. 4 shows the total trip time required by the vehicle to travel from the source to the destination. For the morning and night times, the trip time is the same (around 1.2–1.5 minutes) for both the shortest distance path and data analytics based techniques. This is because both techniques choose route1 for the trip as shown in Fig. 5.

For the afternoon time, we note that the shortest distance path technique results in a trip time of 5.2 minutes due to congestion on the road. In contrast, the data analytics module chooses route2 and results in a trip time of 3.02 minutes. Similarly, for the evening time, the shortest distance path technique requires a trip time of 7.15 minutes due to the accident event. As the data analytics module used only on the past history, it chooses the same route which yields a similar trip time. However, the complete data analytics module that also considers the current vehicle density significantly reduces the trip time by 4 minutes. Therefore, the data analytics module of the TCC can help drivers make efficient decision regarding their route to the destination resulting in a reduction of the trip time.

Based on simulation results obtained, we conclude that keeping a record of centralized traffic density information and applying data analytics provides the highest benefit in case of a sudden event or an accident (6pm time). At other times when the vehicle density is not high, a greedy approach that uses the shortest distance path also provides comparable performance. In addition, for a high density scenario, past vehicle density information yields similar performance to the case when the current vehicle density is used. Therefore, the TCC can reduce the frequency of vehicle density reevaluation (thereby reducing the computational overheads) when no sudden event occurs without degrading the application performance. For city level traffic management, we expect that the generated data will be much larger (compared with that used in our simulation scenario). Therefore, we need to investigate new techniques that can reduce the amount of data disseminated and the frequency of computing vehicle density. One promising solution would be to detect and share information about events (such as accident and congestion) with the data analytics module in order to initiate the re-evaluation of the vehicle density.

4.2. Adaptive transmission range selection

Another important use case of data analytics in C-ITS is to select appropriate wireless transmission parameters of on-board transceivers in order to maximize the application performance. The goal here is to ensure reliable communications between vehicles which cannot be achieved with fixed transmission parameters due

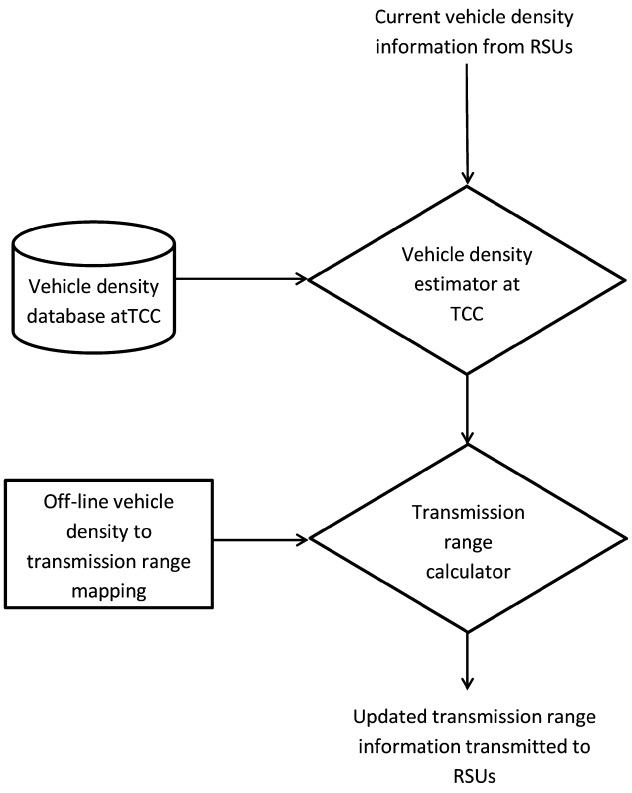


Fig. 6. Transmission range estimator.

Table 4
Off-line mapping of vehicle density and transmission range.

Vehicle Density (vehicles/km)	Transmission Range (m)
0–50	1000
50–100	800
100–150	600
150–200	400
>200	200

to varying vehicle density, thus resulting in large number of packet collisions in a high vehicle density scenario [9].

Fig. 6 presents a mechanism that could be used for adjusting the transmission range of vehicles. As we have explained previously, the TCC continuously estimates the vehicle density at different road sections based on the weighted summation of information in its database and the current mobility information received from RSUs. In addition, the TCC also contains an off-line mapping of relationship between vehicle density and transmission range. This off-line mapping could be developed based on analytical models, large scale simulation studies or real testbed experiments.

Based on this off-line mapping and vehicle density estimate at the TCC, an updated transmission range value is calculated and input into the RSUs. Finally, the RSUs can inform vehicles on a particular road section to adapt their transmission ranges according to vehicle densities. Table 4 presents a simple example of the off-line vehicle density-transmission range mapping. As we mentioned earlier, this mapping should be further enhanced based on extensive empirical measurements and tests.

We implemented the above mechanism in NS-3 simulator and performed simulations using the parameters listed in Table 3. We plot the Packet Delivery Ratio (PDR) of CAMs for vehicles traveling on the road section AD at different times of the day in Fig. 7. When the vehicles use fixed transmission range value of 500 m, the packet delivery ratio varies from 0.62–0.82 depending on time of the day. During morning and night times, the traffic congestion

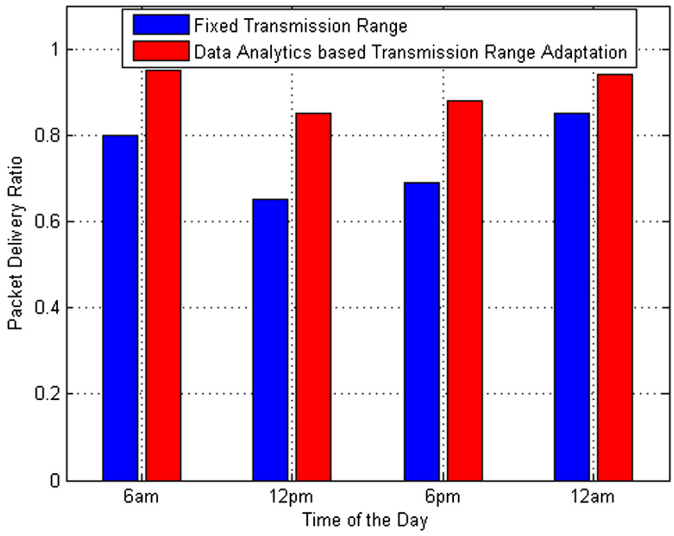


Fig. 7. Packet delivery ratio at different times of the day.

is low, resulting in a higher PDR. During the afternoon time (when school zone time applies) and evening time (when there is an accident on the road), the packet delivery ratio is lower than 0.65.

With the adaptive transmission range technique, vehicles improve their PDR by 0.04–0.2 at different times of the day. The rationale behind this improvement is that during rush hours and traffic congestion, the transmission range is reduced to improve PDR. We note that vehicles could also adaptively control their transmission ranges in a distributed manner without the need of the TCC. This can be achieved with a low complexity and little additional computational cost by using metrics such as the channel busy metric [14]. It is worth noting that this improvement in PDR will also result in increasing the reliability of various safety applications such as collision avoidance. Due to the higher packet reception rate, vehicles will get timely notification of events such as sudden brake applied by a vehicle ahead thereby increasing the level of vehicle safety.

5. Potential future use cases of data analytics in C-ITS

In this section, we discuss some of the potential future use cases of data analytics in C-ITS. These use cases serve as examples where data analytics can have a significant impact on the operational efficiency, robustness, reliability and scalability of C-ITS.

5.1. Traffic management in smart cities

Traffic management will be a key challenge in future smart cities [34]. Since the TCC can access the city level traffic mobility data using all RSUs on the roads, it could play a vital role in managing city traffic efficiently. As an example, the TCC could receive efficient route guidance requests from a large number of vehicles. The data analytics module in the TCC could use variables such as vehicle density, road conditions, weather, travel departure time and so on to make an intelligent routing decision [35]. Instead of optimizing the route for just one vehicle, a complex data analytics function can make decisions to reduce overall traffic congestion on the road. Deep Learning is a promising machine learning technique that can be used for feature and structure identification in a large data set. In the context of C-ITS, deep learning can be used to predict traffic flow information which is an important metric for traffic management [36].

5.2. Enhancing vehicle safety with network security analysis

Since C-ITS applications involve vehicle safety, it is important to ensure the safety of vehicles so that they could take accurate driving decisions. Network security also plays a vital role in C-ITS because ensures that no intruder can inject false messages. The security procedure in C-ITS incurs overhead in terms of packet size and the time required for signing/verifying security signatures [1]. When communications become more secure, the increased security may affect safety awareness because of additional security overheads. Data analytics could play a vital role here by supporting functions such as prioritized verification of more critical packets, signature omission of random packets, and the selection of appropriate security key size and so on [37].

5.3. Data outlier detection

Detection of data outliers due to erroneous sensors are also important to data analytics. There are many sensors mounted on vehicles that provide traffic and mobility information used for safety applications. Any error caused by the malfunctioning of such sensors could prove fatal in the context of C-ITS. Therefore, efficient data outlier detection schemes are required in order to analyze real time data and detect erroneous or attack data when injected into the data stream. For example, we can record a cache of vehicles' positions and speeds. Since vehicles cannot suddenly change their positions and speeds, a spike in data could indicate an error caused by GPS measurements.

6. Conclusion

This paper has explored how data analytics can assist in the area of intelligent transport systems. We presented a detailed C-ITS architecture that includes connectivity between vehicles, RSUs and TCC. We have discussed various technical challenges that arise when data analytics is applied in the context of C-ITS. We conducted simulation tests to demonstrate how data analytics can help C-ITS applications such as efficient route guidance and enhancing wireless communication reliability. Finally, we have highlighted future research opportunities for applying data analytics in C-ITs in areas such as traffic management, adaptive network security, and data forensics.

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